FADIME RENÉE STEMMER

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EDUCATION

MIT/WHOI Joint Program in Oceanography Sep 2023 - Present
PhD in Chemical Oceanography and Marine Biogeochemistry
Department of Earth Atmospheric and Planetary Science, Cambridge, MA
Department of Marine Chemistry and Geochemistry, Woods Hole, MA

MIT/WHOI Joint Program in Oceanography Sep 2023 - Sep 2025
M.Sc. in Chemical Oceanography (GPA: 4.0/4.0)
Department of Earth Atmospheric and Planetary Science, Cambridge, MA
Department of Marine Chemistry and Geochemistry, Woods Hole, MA

Waseda Univerity, Tokyo, Japan Apr 2022 - Sep 2022
B.Sc. in Biosciences (GPA: 3.89/4.0)
Exchange Semester with focus on Biosciences and Japanese language.

Technische Universität München Sep 2019 - Apr 2023
B.Sc. in Biochemistry (GPA: 3.73/4.0)
Fakultät für Chemie, Garching, Germany

RESEARCH EXPERIENCE

Graduate Research Assistant, MIT/WHOI Joint Program in Oceanography
Sep 2023 - Present

- Identified transport proteins involved in the export of proteases and peptidases by marine heterotrophic bacteria for degradation of proteinaceous organic matter.
- Developed an LC-MSMS based method to characterize the cleavage specificity and rate of unknown proteases.
- Characterized the rates of enzymatic protein degradation by marine proteases and heterotrophic communities in culture and in the field.
- *Skills and Tools*: Proteomics, Microbiology, LC-MSMS, UV/Vis Spectroscopy, Fieldwork, Proteome Discoverer, Fragpipe, Scaffold, Skyline, DIA-NN

Graduate Research Assistant, Institute of Structural Biology of Helmholtz Zentrum Munich
Jan 2023 - Aug 2023

- Expression, purification, structural, and biophysical characterization of C19orf12, a protein with unknown function whose mutations cause mitochondrial membrane protein associated neurodegeneration (MPAN).
- *Skills and Tools*: Topspin, MestReNova, Nuclear Magnetic Resonance Spectroscopy, LC-MSMS, SAXS, SANS, X-ray crystallography, Electron Microscopy, UV/Vis Spectroscopy

Undergraduate Researcher, Technical University of Munich, Department of Chemistry, Center for Functional Protein Assemblies (CPA)
Nov 2022 - Dec 2022

- Expression, purification, and structural investigation of the archaeal 20S proteasome, the enzyme LaeA involved in gliotoxin biosynthesis, and t-RNA modifying enzymes.
- *Skills and Tools*: X-ray crystallography, Cryo-Electron Microscopy, Coot

Undergraduate Researcher, Technical University of Munich, Department of Chemistry, Center for Functional Protein Assemblies (CPA)
Aug 2021 - Sep 2021

- Synthesis of promising antibacterial molecules that target the ClpXP complex - a key protein complex for toxin production in bacteria.

· *Skills and Tools*: MestReNova, Nuclear Magnetic Resonance Spectroscopy, LC-MSMS, Organic synthesis.

RESEARCH PUBLICATIONS

Fadime R. Stemmer, Matt R. McIlvin, Mary Ann Moran, Mak A. Saito; "Switch between inorganic and proteinaceous N utilization discovered via secreted (metallo-)proteases in the Model Marine Heterotroph Ruegeria pomeroyi DSS-3. *in Prep.*"

Mariana I. Torres, **Fadime R. Stemmer**, Mak A. Saito, Rogier Braakman; "Purine-mediated cryptic nitrogen cycling among ocean microbes revealed by Metaproteomics-Stable Isotope Probing *in Prep.*"

Fadime R. Stemmer, Matt R. McIlvin, Mak A. Saito; "Characterization of Cleavage Specificity of Novel Proteases in Environmental Microbes *in Prep.*"

Christina Draeger, Joseph Rotondo, Olivia Doty, **Fadime R. Stemmer**, Stella Heflin; "The Graduate Climate Conference as a Model for Equitable Access, a Social Infrastructure, and a Research-Trend Indicator." *in Prep.*"

PRESENTATIONS

Fadime R. Stemmer, Matt R. McIlvin, Mak A. Saito; "Characterization of Cleavage Specificity of Novel Proteases in Environmental Microbes. *2026 Annual Meeting of the American Society of Mass Spectrometry*"

Fadime R. Stemmer, Daniella Asturias, Loay Jabre, Matt R. McIlvin, Mak A. Saito"; Molecular Drivers of Surface Exoprotease Activity Across the Western North Atlantic. *2026 Ocean Sciences Meeting*"

Daniella Asturias, **Fadime R. Stemmer**, Loay Jabre, Matt R. McIlvin, Elizabeth Kujawinski, Mak A. Saito"; Linking Hyperspectral PACE Observations with Underway Metaproteomics along a Northwest Atlantic Transect. *2026 Ocean Sciences Meeting*"

Stella Heflin, Joey Rotondo, Sky Gale, Alex Chang, Christina Draeger, **Fadime R. Stemmer**, Chloe Smith, Ashley Lasinsky, Zilu Meng, Akshay Rajeev; "Building Community and Belonging in Graduate Climate Research: Lessons from the Graduate Climate Conference (GCC). *2025 Annual Meeting of the American Geophysical Union*"

Fadime R. Stemmer, Matt R. McIlvin, Mak A. Saito"; The Exo-proteomic Framework for Nitrogen Acquisition from Proteinaceous Organic Matter in the Model Marine Heterotroph Ruegeria Pomeroyi DSS-3. *2025 Gordon Research Conference in Chemical Oceanography*"

Lina Taenzer, Lukas Taenzer, Kate Lane, Mathieu Dever, Faith Brooks, Thomas Catapano, Margaret Gregory, Jo Hickman, Natalie Hummel, Jacob Knight, Kelby Kramer, Sophie Kuhl, Miah Manning, Sarah Messenger, Andrew Motz, Ellen Mule, Hiroki Nagao, **Fadime R. Stemmer**, Galen Wilcox, Timothy M. Shank"; Updated Snapshot of the Physical and Biogeochemical Conditions Across the Rapidly Warming US Northeast Continental Shelf and Shelfbreak. *2024 Ocean Sciences Meeting*"

FIELD WORK

NSF C-CoMP cruise AE2520 on R/V *Atlantic Explorer*,
Bermuda (BATS) to Woods Hole (NES-LTER). Cruise Report — Sep 2-15, 2025

NSF C-CoMP cruise AE2504 on R/V *Atlantic Explorer*,
Bermuda (BATS) to Woods Hole (NES-LTER). Cruise Report — Mar 9-22, 2025

Student research cruise AR80 on R/V *Neil Armstrong*,
Woods Hole to NES-LTER. — Nov 17-19, 2023

TEACHING EXPERIENCE

Teaching Assistant for 17.925 - MIT Science and Technology Policy Bootcamp Jan 2027 - Jan 2027
MIT, Department of Political Sciences

· One week intensive course on the fundamentals of innovation systems and the economic and technology development factors, and to build a framework to approach science and technology policymaking.

· Curriculum design, invite speakers and panelists, grading essays and student presentations.

Teaching Assistant for 12.384/21H185/STS005 LIVING DANGEROUSLY: Environmental Problems from 1900 to Today Feb 2025 - May 2025
MIT, Department of Earth Atmospheric and Planetary Sciences

· Conducting office hours.
· Correcting student essays.

Teaching Assistant for 12.742 Marine Chemistry Sep 2024 - Dec 2024
MIT, Department of Earth Atmospheric and Planetary Sciences; WHOI, Department of Marine Chemistry and Geochemistry

· Conducting weekly office hours and recitations
· Correcting student assignments and exams.

Academic Tutor for Molecule Structure and Spectroscopy Apr 2022 - Aug 2022; Apr 2023 - Aug 2023;
TUM, Department of Chemistry

· Holding weekly office hours and tutoring sessions.

CERTIFICATIONS

Computational and Statistical Skills for Mass Spectrometry and Proteomics,
Northeastern University issued Jun 2026
Graduate Teaching Development Certificate,
MIT issued Jun 2025
JLPT N2,
The Japan Foundation issued Jan 2025
TOEFL,
ETS Score: 111 issued May 2021 · Expired May 2023
DELF, B1,
Ministère de l'Éducation nationale et de la Jeunesse issued Oct 2016

VOLUNTEERING

Bootcamp Chair, *MIT Science Policy Initiative* Sep 2026 - May 2027

Fundraising Chairperson, *Graduate Climate Conference* Mar 2026 - Present

External Relations Chair, *MIT Science Policy Initiative* Sep 2025 - May 2026

Co-Chair, *Graduate Climate Conference* Mar 2025 - Mar 2026

Advisor, *MIT European Career Fair* Apr 2025 - Mar 2026

Vice President, *MIT European Career Fair* Apr 2024 - Mar 2025

Company Outreach Chair, *MIT European Career Fair* Sep 2023 - Mar 2024

Team Lead Sorbent Development, *TUM Carbon Removal Initiative* Aug 2021 - Aug 2023